MIL-C-60558 (MU)

25 July 1966
SUPERSEDING
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27 December 1962

# MILITARY SPECIFICATION
# CARTRIDGE, INITIATOR, DELAY, XM148

## 1. SCOPE

1.1 This specification covers Cartridge, Initiator, Delay, XM148, for use with propellant actuated devices in aircrew emergency escape systems.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS
Military
- MIL-A-2550 - Ammunition and Special Weapons, General Specification For
- MIL-E-5272 - Environmental Testing, Aeronautical and Associated Equipment, General Specification For
- MIL-H-25279 - Hose, Assembly, Tetrafluorethylene, High Temperature, Power Plant, Aircraft

STANDARDS
Military
- MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
- MIL-STD-109 - Quality Assurance Terms and Definitions
- MIL-STD-414 - Sampling Procedures and Tables for Inspection by Variables for Percent Defective
- MIL-STD-639 - Surface Roughness Values Preferred for Army Ordnance Materiel
- MIL-STD-1167 - Ammunition Data Card

DRAWINGS
Munitions Command
- D10522709 - Cartridge, Initiator, Delay, XM148, Assembly
- C8864168 - Carton, Inner, Packing, Ammunition, for Cartridge, Initiator, Delay, XM148
- C8864169 - Carton, Outer, Packing, Ammunition, for Cartridge, Initiator, Delay, XM148

FSC 1377

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2.1 (cont)

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Materials, parts, assemblies, chemical and physical requirements shall conform to the applicable drawings, referenced specifications and the following:

3.2 Initial production sample.- Unless otherwise directed by the contracting officer, an initial production sample shall be required.

3.3 Protective finish.- When a protective finish is required by component details, finish requirements shall be as required on the drawings and referenced specifications and shall apply to all surfaces of the assembled cartridge regardless of contour configuration.

3.4 Delay element.- The Delay Element, Cartridge, XM44, shall conform to the applicable drawing and referenced specification.

3.5 Cartridge assembly.- Components shall be combined to produce an assembly which conforms to the requirements of drawing D10522709.

3.5.1 Vibration.- The assembled cartridge shall withstand vibration as prescribed in MIL-E-5272, Procedure XII, without evidence of any disintegration of the delay charge, primer, igniter or propellant, or derangement of components.

3.6 Functional.

3.6.1 The cartridge shall not fail to fire.

3.6.2 Powder charge.- The igniter and propellant charges shall be fixed by the contractor to produce the specified ballistic requirements.

3.6.2.1 Primer sensitivity.- The primer shall fire when drop tested at an energy level of 26 inch ounces. A sectioned cartridge, void of the charged delay body, igniter and propellant charge, shall be used for this test.

3.6.3 Function.- The cartridge, when tested in an XM64 or XM76 Initiator shall produce the pressures and delay times at the temperatures specified in table I. Pressure values shall be recorded at the end of three feet of size No. 4 aircraft hose which complies with MIL-H-25279 and a pressure chamber with a volume of 1 cubic inch.



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Table I

| Temperature (° F) | Pressure (psi) | | Delay (seconds) |
|---|---|---|---|
| | min | max | |
| -65 to -70 | 1300 | 2200 | 0.90 max |
| 65 to 75 | 1500 | 2200 | 0.75 ± 0.10 |
| 160 to 165 | 1500 | 2200 | 0.60 min |

3.7 Workmanship.- Workmanship shall be in accordance with the requirements of the applicable drawings, referenced specifications and the following:

3.7.1 Metal defects.-All metal parts shall be free from cracks, splits, laminations, perforations, porosity or any similar defect.

3.7.2 Surface finish.- The requirements for surface finish are as detailed by the applicable drawing and referenced standard.

3.7.3 Burr.- No part shall have a burr which might interfere with the assembly or function of the item, or which might be injurious to personnel handling the item.

3.7.4 Foreign matter.- No part or assembly shall contain chips, dirt, grease, oil, rust, corrosion or any other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550.

4.2 Initial production sample.- At the beginning of regular production, the contractor shall submit an initial production sample to a Government approved facility for evaluation. The sample shall consist of 80 cartridges and 10 sets of unassembled components, including explosive components. The sample shall be manufactured in the same manner, using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing, must be the same as will be used in regular production and shall be obtained from the same



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4.2 (cont)

source of supply. After inspection and provisional acceptance at source, the initial production sample shall be inspected at the Government approved facility for all requirements of the drawings and specifications. The Government reserves the right to require new initial production samples until such time as an acceptable sample is submitted.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Presentation of lot.- The product shall be presented in accordance with MIL-STD-105 and MIL-STD-414, as applicable.

4.3.1.2 Size of lot.- Unless otherwise specified by the procuring agency an ammunition lot shall consist of 600 cartridges, maximum, exclusive of test samples, and shall consist of:

a. Components produced by a homogeneous process.

b. Propellant from not more than one lot.

c. Igniter from not more than one lot.

d. Delay elements from not more than one lot.

4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects, using the sampling plan and criteria in tables II and III. For an individual defect (single characteristic), the criteria in table II shall be used. For total defects within a major or minor class, the criteria in table III shall be used. All nonconforming material shall be rejected. The Ordnance Standard Finish Blocks prescribed in MIL-STD-639 shall be used as a basis of comparison for surface roughness determination. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

Table II

| Lot Size | Sample Size | Acceptance Numbers Major | Minor |
|---|---|---|---|
| 1 - 300 | 100% examination | 0 | 0 |
| 301 - 1300 | 110 | 1 | 2 |
| 1301 - 3200 | 150 | 2 | 3 |



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4.3.2 (cont)

Table III

| Number of Defects in Class | Total Major Defects Allowable |
|---|---|
| 2 - 3 | 1 |
| 4 - 7 | 2 |
| 8 - 19 | 3 |

| Number of Defects in Class | Total Minor Defects Allowable |
|---|---|
| 2 - 3 | 2 |
| 4 - 7 | 3 |
| 8 - 16 | 4 |

4.3.2.1 Classification of defects.

4.3.2.1.1 Cartridge, Initiator, Delay, XM148 Assembly, Dwg D10522709.

| Defect | Method of Inspection |
|---|---|
| CRITICAL | |
| 1. Any part missing | X-ray/Visual 1/ |
| 2. Improper assembly | X-ray/Visual 1/ |
| 3. Mixed ammunition | Visual |
| 4. Metal defective | Visual |
| 5. Crimp missing or incorrect | Visual |
| 6. Marking missing, incorrect or illegible | Visual |
| MAJOR | |
| 101. Cartridge profile failure | Gage |
| 102. Diameter of body, max | Gage |
| 103. Diameter of head, max | Gage |
| 104. Length to datum diameter, max | Gage |
| 105. Diameter of head recess, min | Gage |
| 106. Depth to face of head | Gage |
| 107. Firing pin or retaining cup above flush with face of head | Gage |
| 108. Burr | Visual |
| 109. Foreign matter | Visual |
| 110. O-ring split, cracked or cut | Visual |
| MINOR | |
| 201. Total length | Measure |
| 202. Dent | Visual |
| 203. Draw scratch | Visual |
| 204. Fold, wrinkle, buckle or bulge | Visual |
| 205. Varnish inadequate or missing | Visual |
| 206. Surface finish improper | Visual |

1/ The assembled cartridge shall be x-rayed with the long axis of the cartridge perpendicular to the rays of the x-ray machine. The x-ray negatives, properly identified, shall be forwarded to the procuring agency.



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4.3.2.1.2 Case, Cartridge, Dwg C10519996 covering a detail of Dwg D10522709.

| Defect | | Method of Inspection |
|---|---|---|
| CRITICAL | | |
| 1. | Metal defective | |
| MAJOR | | |
| 101. | Rockwell hardness incorrect | Gage |
| 102. | Large diameter, max | Gage |
| 103. | Small diameter, max | Gage |
| 104. | Base wall thickness | Gage |
| 105. | Wall thickness | Gage |
| 106. | Length to datum diameter | Gage |
| 107. | Concentricity of large and small diameters | Gage |
| MINOR | | |
| 201. | Total length, max | Gage |
| 202. | Fold, wrinkle, buckle or bulge | Visual |
| 203. | Surface finish improper | Visual |
| 204. | Burr | Visual |
| 205. | Overall length | Gage |
| 206. | Radii incorrect | Gage/Visual |
| 207. | Draw scratch | Visual |

4.3.2.1.3 Carton, Packing, Ammunition, for Cartridge, Initiator, Delay, XM148, Dwg C8864168.

CRITICAL
None defined.

MAJOR
None defined.

| | | |
|---|---|---|
| MINOR | | |
| 201. | Contents loose | Manual/Visual |
| 202. | Quantity of contents incorrect | Visual |
| 203. | Separator or filler missing | Visual |
| 204. | Heat sealer incomplete, loose or distorted | Visual |
| 205. | Cut or damaged through impregnated box | Visual |
| 206. | Marking missing, incorrect or illegible | Visual |

4.3.2.1.4 Box, Packing, Ammunition, for Cartridge, Initiator, Delay, XM148, Dwg C8864169.

CRITICAL
None defined.

MAJOR
None defined.



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4.3.2.1.4 (cont)

| | Defect | Method of Inspection |
|---|---|---|
| MINOR | | |
| 201. | Contents loose | Manual/Visual |
| 202. | Hardware missing, loose, broken or improperly engaged | Visual |
| 203. | Strapping missing, loose, broken or improperly assembled | Visual |
| 204. | Contents of box exposed | Visual |
| 205. | Box damaged | Visual |
| 206. | Marking missing, incorrect or illegible | Visual |
| 207. | Number of cartons incorrect | Visual |
| 208. | Car seal missing, unsealed or improperly positioned | Visual |

4.3.3 Delay element examination.- Prior to assembly into a cartridge, each XM44 Delay Element shall be visually examined for presence of primer and delay charge or for deformation of any kind. Any deformation or absence of the primer or delay charge shall be cause for rejection of the delay element.

4.3.4 Inspection testing.- The following tests shall be performed in accordance with the provisions of 4.4:

4.3.4.1 Protective finish.- The number of units used and the acceptance criteria for the protective finish tests shall be in accordance with the applicable specification. Failure of any sample to comply with the requirements of the applicable specification shall be cause for rejection of the lot.

4.3.4.2 Airtightness.- Each cartridge shall be subjected to the specified pressure differential. Failure to comply with the requirements of the drawing shall be cause for rejection of the cartridge.

4.3.4.3 Vibration test, initial production sample.- Twelve of the 30 assembled, x-rayed cartridges of the initial production sample, properly identified, shall be subjected to vibration as directed in 3.5.1. After the vibration test, the 12 vibrated cartridges shall be x-rayed as directed in Note 1/ of 4.3.2.1.1. Evidence of any disintegration of the primer, delay charge, propellant or igniter, or derangement of any component shall be cause for rejection of the lot.

4.3.4.4 Vibration test, production lot sample.- When it becomes necessary to make a change in the lot of delay elements, propellant or igniter after production is initiated, 30 cartridges from the first production lot thereafter, shall be tested in accordance with 4.3.4.3.

4.3.4.5 Primer sensitivity.- A sample of 50 assembled cartridges from the initial production sample, and 25 assembled cartridges from each subsequent production lot shall be tested for sensitivity. Failure of the sample to comply with the requirement of 3.6.2.1 shall be cause for rejection of the lot.



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4.3.4.6 Functioning.- The 30 test cartridges, including the 12 vibrated cartridges, of the initial production sample (or subsequent sample of 30 cartridges when a change has been made in the lot of delay elements, propellant, igniter or primers) and 21 cartridges from each subsequent production lot, shall be temperature conditioned as prescribed in table IV and 4.4.5.1 and then tested in accordance with 4.4.5.2. Four of the 12 vibrated cartridges shall be included in each test group. The variables sampling plan of MIL-STD-414, example C2 or C3, as applicable, for the AQL listed in table IV shall be used for each group. Failure of any cartridge to comply with the requirement of 3.6 or failure of a group to meet the criteria of the variables sampling plan, shall be cause for rejection of the lot.

Table IV

| Temperature Degrees (°) F | Sample Size | | AQL Percent |
|---|---|---|---|
| | Initial Production Lot | Production Lot | |
| -65° to -70° | 10 | 7 | 0.65 |
| 65° to 75° | 10 | 7 | 0.65 |
| 160° to 165° | 10 | 7 | 0.65 |

4.3.5 Inspection equipment.- The examination and tests shall be made using equipment approved by the Government.

4.4 Test methods and procedures.

4.4.1 Protective finish.- Test procedures for protective finish shall be in accordance with the applicable military standard referenced on the drawing.

4.4.2 Airtightness.- The method and procedure for determining airtightness shall be by use of equipment approved by the procuring agency.

4.4.3 Vibration.- The method and procedure for performing the vibration test shall be in accordance with MIL-E-5272, Procedure XII.

4.4.4 Primer sensitivity.- Remove and discard the charged delay body and all powder charges from the test cartridges by sectioning the cartridge case around its periphery 1 inch, maximum, above the base. Drop test 50 sectioned cartridges from the initial production sample and 25 sectioned cartridges from each production lot (all void of charged delay body and powder charges) at 13 inches, using a steel ball 15/16 inch in diameter, weighing 1.94 ounces plus or minus 0.02 ounce, in a primer sensitivity machine with a cartridge holder approved by the procuring agency.

4.4.5 Functional tests.- The following tests will be performed by the Government:

4.4.5.1 Temperature conditioning.- Cartridges will be loaded into an acceptable XM64 or XM76 Initiator, with end caps to cover the gas port during the conditioning period. The temperature of the propellant within the cartridge will be considered as the conditioning temperature and will be determined by means of



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a thermocouple attached to the center of the propellant charge of specimen cartridge assembled in a device similar to the initiator undergoing tests. The conditioning time will be a minimum of 1 hour longer than the time necessary to bring the element contained in the specimen cartridge to the required temperature.

4.4.5.2 Test procedures.- Upon completion of the required temperature conditioning the cartridges will be ballistically tested in accordance with "Frankford Arsenal Manual of Functional Tests for Cartridge Actuated Devices for Aircraft Ejection Systems" (or such method as is approved by the Government) within five minutes after removal from the conditioning chamber. The cartridges will be re-conditioned if this time is exceeded. When re-conditioning cartridges at minus 65 degrees to minus 70 degrees F, all condensation shall be removed from the interior surface of the initiator before the initiator is returned to the conditioning chamber.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.- Packing and marking shall be in accordance with drawings C8864168 and C8864169 or as required by contract.

5.2 Data cards.- Data card information shall be as specified in MIL-STD-1167.

6. NOTES

6.1 Delay time.- Delay time is defined as the time that elapses between the firing of the primer and the beginning of a continuous pressure rise on a pressure record.

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# SPECIFICATION ANALYSIS SHEET

Form Approved
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INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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